

Dual DC-DC Converter Board

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Design, Simulation, Component Selection, Schematic Capture, PCB Layout, BOM Generation, and Verification
Using LTspice and Altium Designer

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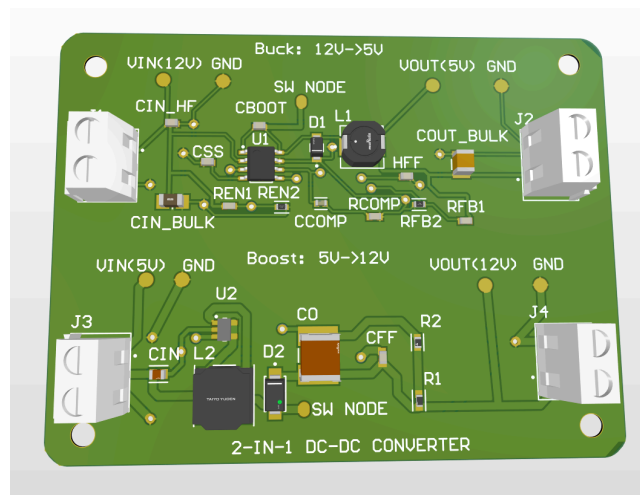


Figure 1. Final 3D PCB Rendering in Altium Designer

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Abstract- *This project presents the design, simulation, schematic capture, component selection, printed circuit board (PCB) layout, and verification of a custom 2-In-1 DC-DC Converter Board developed using LTspice and Altium Designer.*

The system integrates two independent switching power supplies on a single PCB: a buck converter that steps 12 VDC down to 5 VDC and a boost converter that steps 5 VDC up to 12 VDC. The design process included converter topology selection, engineering calculations, component sizing, manufacturer part selection, simulation-based validation, schematic development, PCB layout implementation, design rule verification, and bill of materials (BOM) generation. Particular emphasis was placed on switching regulator design practices, power stage component selection, PCB routing considerations, and manufacturability.

A two-layer PCB was developed and verified through Altium Designer design rule checks. LTspice simulations were performed to validate converter operation, output regulation, and switching behavior prior to PCB implementation. The completed design demonstrates a complete hardware development workflow from initial concept through verified PCB design while applying fundamental principles of power electronics, circuit design, and PCB development.

1. Project Overview

1.1 Project Summary

The objective of this project was to design and validate a custom dual-function DC-DC converter board capable of both step-down and step-up power conversion on a single printed circuit board. The design integrates two independent power conversion circuits: *Buck Converter: 12V input to regulated 5V output & Boost Converter: 5V input to regulated 12V output.*

The project was completed using LTspice for circuit simulation and Altium Designer for schematic capture, component selection, PCB layout, design rule verification, and bill of materials generation. The primary goal was to demonstrate a complete hardware development workflow from concept generation through PCB implementation while applying power electronics design principles and PCB layout best practices.

1.2 Design Objectives

The following objectives were established at the start of the project:

Functional Objectives

- Convert 12V DC to 5V DC using a switching buck regulator
- Convert 5V DC to 12V DC using a switching boost regulator
- Maintain stable regulated output voltages
- Validate converter operation through simulation

Hardware Design Objectives

- Create production-style schematics using manufacturer components
- Generate a complete Bill of Materials (BOM)
- Implement a two-layer PCB layout
- Incorporate ground plane implementation
- Include mounting provisions for mechanical integration
- Pass design rule verification without violations

Professional Development Objectives

- Demonstrate component selection methodology
- Practice PCB layout techniques for switching regulators
- Document engineering tradeoffs and design decisions
- Produce portfolio-quality engineering documentation

Table 1. Design Requirements

Parameter	Input Voltage	Output Voltage	Converter Type	Switching Topology
Buck Converter	12VDC	5VDC	Step-Down	Buck
Boost Converter	5VDC	12VDC	Step-Up	Boost

1.3 System Architecture

The final PCB contains two electrically independent power conversion circuits. Both converters were implemented on a common two-layer PCB while maintaining independent power paths, allowing each converter to be evaluated and validated separately.

Buck Converter Section

Input: 12V DC
Output: 5V DC
Controller: TPS54331D Buck Regulator
Primary Components:

- TPS54331D Controller
- Power Inductor
- Schottky Diode
- Input Capacitors
- Output Capacitors
- Feedback Network

Boost Converter Section

Input: 5V DC
Output: 12V DC
Controller: TPS61040DBVR
Primary Components:

- TPS61040 Boost Controller
- Power Inductor
- Schottky Diode
- Output Capacitor
- Feedback Network

1.4 Design Workflow Overview

This project demonstrates practical experience in:

1. Power Electronics Design
2. Switching Converter Theory
3. LTspice Simulation
4. Component Selection
5. Schematic Capture
6. PCB Layout
7. Ground Plane Design
8. Design Rule Verification (DRC)
9. BOM Creation
10. Engineering Documentation

2. DC-DC Converter Fundamentals

2.1 Introduction to Switching Regulators

DC-DC converters are electronic circuits used to efficiently convert one DC voltage level

into another. Unlike linear regulators, which dissipate excess energy as heat, switching regulators transfer energy through controlled switching action, inductive energy storage, and capacitive filtering. This results in significantly higher power conversion efficiency and makes switching converters suitable for battery-powered systems, embedded electronics, industrial controls, and power distribution applications. This project incorporates two common switching converter topologies: a buck converter for voltage step-down operation and a boost converter for voltage step-up operation.

2.2 Buck Converter Fundamentals

A buck converter is a switching regulator used to reduce an input voltage to a lower output voltage.

In this project, the buck converter converts: **12 VDC $\rightarrow$ 5 VDC**

The converter operates by rapidly switching a transistor on and off at a fixed frequency. During the ON state, energy is transferred from the input supply into the inductor. During the OFF state, the inductor releases stored energy to the load through the freewheeling diode. The output capacitor smooths the resulting voltage ripple to provide a stable DC output.

Buck Converter Operating States

Switch ON

- Current flows from the input source through the inductor.
- Inductor current increases.
- Energy is stored in the magnetic field of the inductor.
- Output capacitor supplies load current and charges.

Switch OFF

- Stored inductor energy continues supplying the load.
- Current flows through the Schottky diode.
- Inductor current decreases.
- Output capacitor helps maintain output regulation.

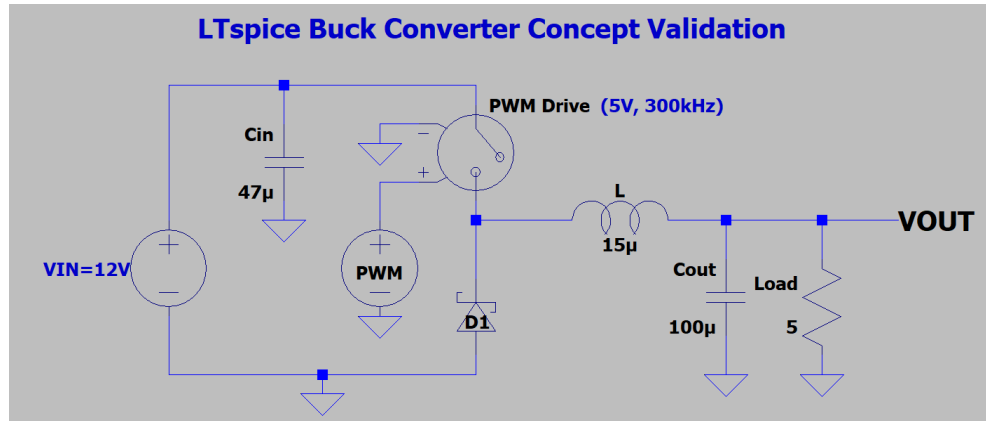


Figure 2. Conceptual Buck Converter Model Used for LTspice Validation

2.3 Boost Converter Fundamentals

A boost converter is a switching regulator used to increase an input voltage to a higher output voltage.

In this project, the boost converter converts: **5 VDC $\rightarrow$ 12 VDC**

The converter operates by storing energy within an inductor during the switch ON period and releasing that energy to the output during the OFF period. The resulting output voltage exceeds the input voltage because the inductor-generated voltage is added to the source voltage during the energy transfer process.

Boost Converter Operating States

Switch ON

- Current flows through the inductor.
- Inductor current increases.
- Energy is stored in the magnetic field of the inductor.
- Output capacitor supplies load current.

Switch OFF

- Inductor voltage reverses polarity.
- Stored energy is released to the output.
- Current flows through the Schottky diode.
- Output capacitor recharges and maintains output regulation.

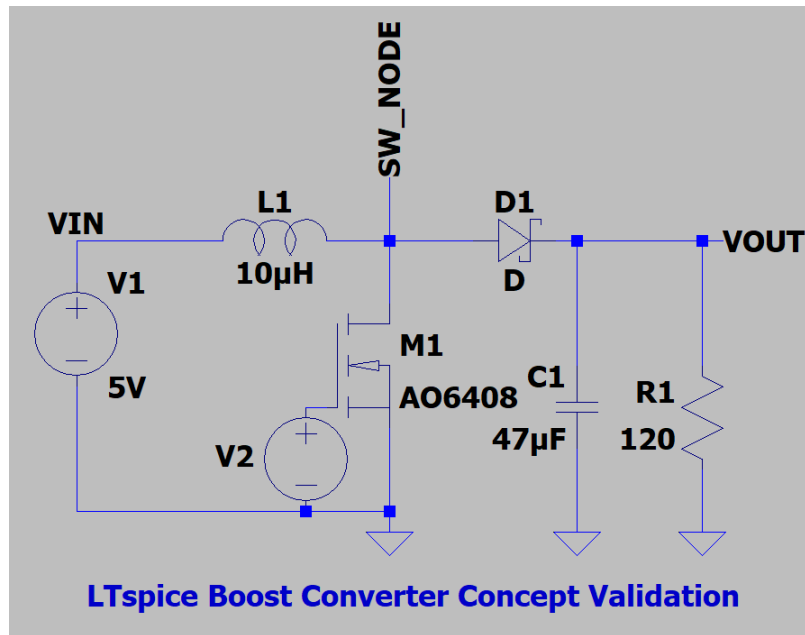


Figure 3. Conceptual Boost Converter Model Used for LTspice Validation

2.4 Power Conversion Principles

Both converters rely on three fundamental energy-storage components:

1. **Inductor-** The inductor stores energy in a magnetic field while current increases and releases energy when current decreases. Inductor selection directly affects current ripple, transient response, and efficiency.
2. **Capacitor-** Capacitors store electrical energy and reduce output voltage ripple. Input capacitors stabilize the source voltage while output capacitors smooth switching-related fluctuations.
3. **Diode-** Schottky diodes are commonly used in switching converters due to their low forward voltage drop and fast switching characteristics, improving efficiency and reducing power loss.

3. Engineering Calculations

3.1 Buck Converter Duty Cycle

The initial calculations began with an *Ideal duty cycle* for a Buck Converter is $D = \frac{V_{out}}{V_{in}}$. Having established $V_{in} = 12V$ & $V_{out} = 5V$, we substituted our values $D = \frac{5}{12} = 41.7\%$ resulting in a calculated value of 41.7% Duty Cycle.

Real-world components introduce conduction and switching losses that reduce overall converter efficiency. To account for these effects, a practical efficiency of 90% was assumed during preliminary design calculations. Substituting that into $D = \frac{V_{out}}{V_{in(max)} \times \eta} \rightarrow D = \frac{5}{12 \times 0.90} = 46.3\%$, which resulted in a slightly higher practical Duty Cycle of 46.3%.

The calculated duty cycle of approximately 46% confirms that the selected buck topology can achieve the required 12 V to 5 V conversion while operating within a practical switching range. This calculation served as an initial validation point prior to LTspice simulation and component selection.

Design Note: The final TPS54331D controller internally regulates duty cycle through its feedback loop. The calculations presented here were used for topology validation and preliminary component sizing rather than direct PWM implementation.

3.2 Buck Converter Inductor Selection

The buck converter inductor was selected to limit ripple current while maintaining stable energy transfer between the switching regulator and the output load. In a buck converter, the inductor stores energy during the switch ON time and releases energy during the switch OFF time. As part of the preliminary component sizing process, the inductor ripple current was estimated using the standard buck converter ripple-current equation $\Delta I_L = \frac{(V_{in}-V_{out}) \times D}{L \times f_s}$. Substituting in $V_{in} = 12V$, $V_{out} = 5V$, $D = 41.7\%$, $L = 15\mu H$, $f_s = 570kHz$ to the formula, $\Delta I_L = \frac{(12V-5V) \times (417)}{15\mu H \times 570kHz}$, resulted in an approximate ripple current of 341mA.

In order to prevent core saturation during normal operation, the selected inductor should have a saturation current rating greater than 3.17A.

$$I_L(peak) = I_{out(max)} + \frac{\Delta I_L}{2} \rightarrow I_L(peak) = 3A + \frac{341mA}{2} = 3.17A$$

Having substituted in the established boundaries, it was confirmed that

$I_L(peak) = 3.17A > I_{out(max)} = 3A$. Therefore, the selected inductor should have a saturation current rating greater than 3.17 A to ensure reliable operation without magnetic core saturation under maximum load conditions.

Design Note: The calculated ripple current of approximately 341 mA represents about 11.4% of the TPS54331D maximum 3 A output capability. A 15 μH inductor was selected because it provides a practical balance between current ripple, transient response, physical size, and PCB layout constraints while remaining within the operating recommendations of the TPS54331D. A larger inductance would further reduce ripple current but increase component size and transient response time. A smaller inductance would reduce board area but increase ripple current and peak current stress. The selected 15 μH value provided an acceptable compromise between electrical performance and PCB implementation constraints. The selected inductor value was subsequently verified during LTspice simulation to ensure acceptable current ripple and stable converter operation under the specified operating conditions.

3.3 Buck Converter Capacitor Selection

The output capacitor was selected to reduce output voltage ripple and provide energy storage during switching transitions. In a buck converter, the output capacitor works together with the inductor to smooth the switching waveform into a regulated DC output voltage. The initial design utilized a 22 μF ceramic output capacitor based on preliminary sizing calculations. During LTspice validation, the converter exhibited greater transient deviation and output ripple than desired during startup and load regulation conditions. To improve output voltage stability, the capacitor value was increased to 47 μF . Having $\Delta V_{out} = \frac{\Delta I_L}{8 \times f_s \times C_{out}}$ where $\Delta I_L = 341mA$, $f_s = 570kHz$, and $C_{out} = 47\mu F$ we substituted in the values. $\Delta V_{out} = \frac{341mA}{8 \times 570kHz \times 47\mu F}$ resulting in $\Delta V_{out} = 1.59mV$. The calculated ripple voltage of approximately 1.59 mV indicates that the selected 47 μF output capacitor provides sufficient filtering to maintain a stable 5 V output while minimizing switching-induced voltage fluctuations. Increasing capacitance beyond 47 μF would further reduce ripple but at the expense of increased component size and cost.

Design Note: The final design utilized a 47 μF ceramic output capacitor due to its low ESR, compact footprint, and excellent high-frequency filtering characteristics. While a larger capacitor would further reduce output ripple, it would also increase component size and cost. The selected 47 μF value provided an appropriate balance between output voltage stability, transient response, PCB area utilization, manufacturability, and overall design complexity. The final capacitor value was subsequently validated through LTspice simulation to verify startup behavior, output regulation, and steady-state performance.

3.4 Boost Converter Duty Cycle

The initial calculations for the boost converter began with the ideal boost converter duty cycle equation. Unlike a buck converter, a boost converter increases the output voltage above the input voltage by storing energy in an inductor during the switch ON period and releasing that energy to the output during the switch OFF period. For an ideal boost converter duty cycle $D = 1 - \frac{V_{in}}{V_{out}}$. Having established $V_{in} = 5V$ & $V_{out} = 12V$, we substituted the values in $D = 1 - \frac{5V}{12V}$, resulting in a duty cycle of 58.3%. However real-world converters experience conduction losses, switching losses, diode losses, and other non-ideal effects that reduce overall efficiency. For preliminary design calculations, an efficiency of 90% was assumed. Substituting those values resulted in a practical duty cycle of 62.5%, $D = 1 - \frac{5V \times 0.90}{12V} = 62.5\%$. The calculated duty cycle indicates that the switching device must remain ON for approximately 62% of each switching cycle to achieve the required 12 V output from a 5 V input source. The higher duty cycle relative to the buck converter reflects the increased energy transfer required to boost the input voltage from 5 V to 12 V. This value was subsequently used during inductor and output capacitor sizing calculations.

Design Note: The final TPS61040DBVR boost controller internally regulates duty cycle through its feedback network and control loop. The calculations presented here were used for topology validation and preliminary component sizing rather than direct PWM implementation.

3.5 Boost Converter Inductor Selection

The boost converter inductor was selected to provide sufficient energy storage during the switch ON period while maintaining acceptable current ripple and transient performance. In a boost converter, the inductor stores energy when the switching device is ON and releases that energy to the output during the OFF period. The selected inductor value directly influences ripple current, efficiency, peak current stress, and overall converter performance.

A 10 μH inductor was selected for the final design based on the TPS61040DBVRG4 design recommendations and the target conversion of 5 V to 12 V. Smaller inductance values can increase ripple current and peak current stress, while larger inductance values reduce ripple current at the expense of slower transient response and increased component size. The selected inductor was required to support the peak current demands of the boost converter without entering magnetic saturation. Therefore, the saturation current rating of the selected inductor was reviewed to ensure reliable operation under maximum load conditions.

Design Note: The final design utilized a 10 μH shielded power inductor (Taiyo Yuden NRS8040T100MJGJ). This value provided an appropriate balance between energy storage capability, transient response, physical size, and PCB implementation constraints. The selected component also offers magnetic shielding, which helps reduce electromagnetic interference (EMI) and improves overall converter robustness.

3.6 Boost Converter Capacitor Selection

The output capacitor was selected to reduce output voltage ripple and provide energy storage during switching transitions. In a boost converter, the output capacitor plays a critical role in maintaining output regulation because it supplies load current while the switch is ON and recharges during the OFF period when energy is transferred through the diode to the output. During LTspice validation, the initial 22 μF capacitor resulted in greater output voltage deviation during startup and load transitions. Increasing the output capacitance to 47 μF improved output stability and reduced transient voltage variation. The larger capacitance provided additional energy storage during switching events and improved regulation performance.

Design Note: The final design utilized a 47 μF ceramic output capacitor due to its improved transient performance, low ESR characteristics, and ability to provide stable output regulation during startup and steady-state operation. While larger capacitance values could further reduce output ripple, they would increase component size and cost. The selected 47 μF value provided an effective balance between electrical performance, PCB area, and manufacturability.

3.7 Feedback Network Design

The feedback network was designed to regulate the converter output voltage by providing a scaled representation of the output voltage to the controller's feedback pin. The controller continuously compares the feedback voltage to its internal reference voltage and automatically adjusts the switching duty cycle to maintain a stable regulated output. A resistor divider network was implemented between the output voltage and ground. The midpoint of the divider was connected to the feedback (FB) pin of the switching regulator.

Buck Converter Feedback Network: For the TPS54331D buck converter, the feedback voltage is regulated to an internal reference voltage of approximately 0.8 V. The output voltage is determined by: $V_{out} = V_{ref}(1 + \frac{R_{top}}{R_{bottom}})$

We established that $V_{ref} = 0.8\text{V}$, $R_{top} = \text{Upper Feedback Resistor}$, $R_{bottom} = \text{Lower Feedback Resistor}$.

Substituting the design values, $V_{out} = 0.8\text{V}(1 + \frac{52.3\text{k}\Omega}{10\text{k}\Omega})$ which resulted in $V_{out} = 4.98\text{V}$, which is within 0.4% of the target voltage of 5V. The calculated output voltage of approximately 5 V confirms that the selected resistor divider values satisfy the design requirement for the buck converter output.

Boost Converter Feedback Network: The TPS61040 boost converter regulates its feedback pin to an internal reference voltage of 1.233 V. The output voltage is determined by the resistor divider formed by R1 and R2. Substituting in the values, $V_{out} = 1.233\text{V}(1 + \frac{88.7\text{k}\Omega}{10\text{k}\Omega})$ resulted in $V_{out} = 12.17\text{V}$ which is within 1.4% of the target voltage of 12V. The calculated output voltage of approximately 12 V confirms that the selected feedback network satisfies the boost converter design objective. The calculated feedback network voltages closely matched the desired converter outputs, confirming that the selected resistor values would allow both converters to achieve their target regulated output voltages prior to simulation validation.

Design Note: The resistor divider networks were selected to produce output voltages as close as possible to the target values while using standard resistor values available from manufacturer component libraries. Final output regulation was subsequently verified through LTspice simulation and schematic implementation. Small deviations from the target voltage are expected due to resistor tolerances, component variation, and converter operating conditions.

4. Key Component Selection

Table 2. Key Component Selection Summary

Component	Part #	Function	Selection Rationale
Buck Controller	TPS54331D	12V→5V Regulation	3A output capability, integrated MOSFET, 570kHz switching frequency, wide input range
Boost Controller	TPS61040DBVRG4	5V→12V Regulation	Simple boost architecture, low component count, suitable output capability
Buck Inductor	Murata A921CY-150M-P3	Energy Storage	15μH value provided acceptable ripple current, sufficient saturation current rating
Boost Inductor	Taiyo Yuden NRS8040T100MJGJ	Energy Storage	10μH value matched TPS61040 design recommendations and transient requirements
Buck Output Capacitor	47μF Ceramic	Output Filtering	Improved startup and transient response compared to initial 22μF design
Boost Output Capacitor	47μF Ceramic	Output Filtering	Reduced output voltage deviation and improved regulation stability
Buck Diode	NRVBSS34FA	Freewheeling Path	Schottky diode selected for low forward voltage and fast switching
Boost Diode	SS14	Output Rectification	Schottky diode minimizes losses and improves efficiency
Connectors	TE Connectivity 282836-2	Power Interface	Simple, low-cost power input/output connections suitable for prototype validation

The components shown in Table 2 were selected based on electrical performance requirements, manufacturer recommendations, component availability, PCB implementation constraints, and LTspice simulation results. Particular emphasis was placed on converter efficiency, transient response, output regulation, and manufacturability. Several component values, including the output capacitors, were refined throughout the design process based on simulation observations and design tradeoff analysis.

5. LTspice Validation

LTspice simulations were performed to validate converter operation before PCB implementation. The simulations were used to observe startup behavior, switching activity, output regulation, and transient response for both the buck and boost converter sections.

5.1 Buck Converter Output Voltage Validation

The buck converter simulation was used to verify the 12 V to 5 V step-down conversion. During startup, the output voltage initially overshoot before settling toward its regulated operating point. This response helped evaluate the output capacitor selection, compensation behavior, and overall converter stability.

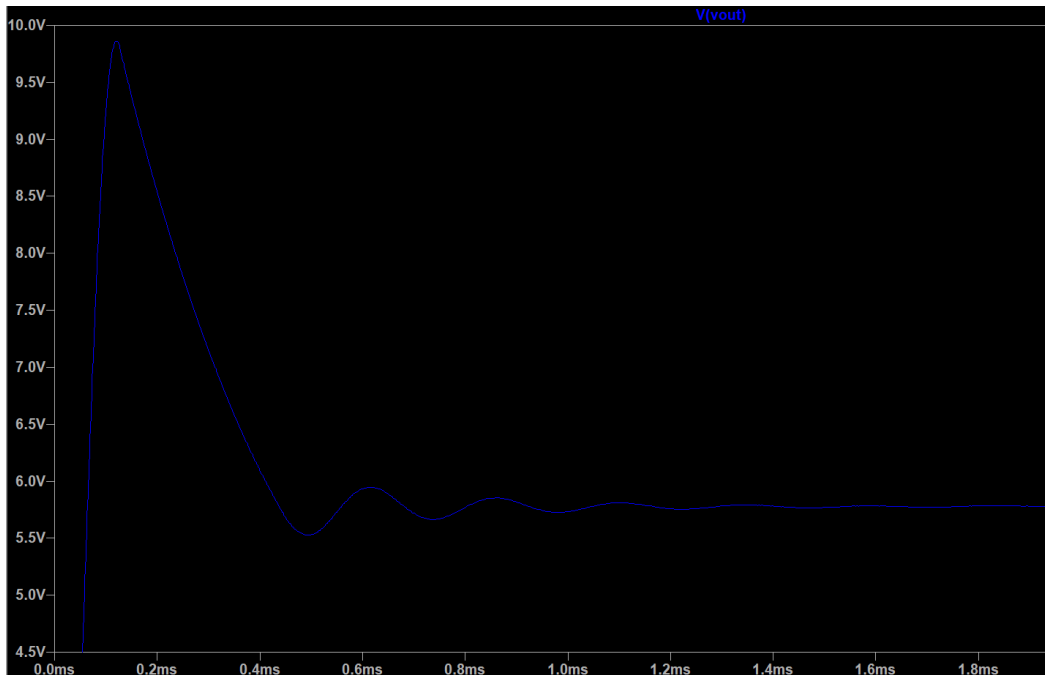


Figure 4. Buck Converter Startup and Transient Response

The waveform demonstrates the output voltage response during startup. The converter transitions from the initial startup condition toward a stable regulated output. The observed transient behavior was used to evaluate the power stage response and confirm that the converter was operating according to buck converter principles.

5.2 Buck Converter Switching Node Validation

The buck converter switching node was monitored to verify proper energy-transfer behavior. In a buck converter, the switching node rapidly transitions as the inductor stores energy during the switch ON period and transfers energy to the output during the switch OFF period.

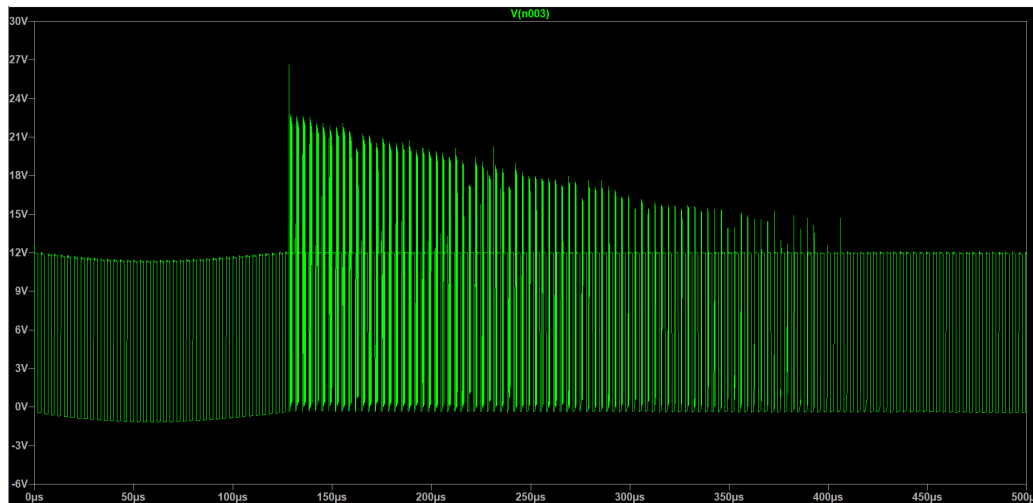


Figure 5. Buck Converter Switching Node Voltage

The waveform shows high-frequency switching behavior at the buck converter switching node (PH). The voltage transitions between the input voltage and ground as the internal switching transistor turns ON and OFF. These switching events confirm proper energy transfer through the inductor and freewheeling diode while regulating the output from 12 V to 5 V. The observed waveform validates expected buck converter operation and confirms correct interaction between the switching regulator.

5.3 Boost Converter Output Voltage Validation

The boost converter simulation was performed to verify the ability of the selected topology to increase the input voltage from 5 V to the desired 12 V output. LTspice was used to evaluate startup behavior, output regulation, and switching operation prior to schematic implementation and PCB layout.



Figure 6. Boost Converter Startup and Regulation Response

The simulated waveform demonstrates the output voltage increasing from the 5 V input source toward the regulated 12 V output. During startup, the converter stores energy within the inductor during the switch ON period and transfers that energy to the output through the Schottky diode during the switch OFF period. The resulting output voltage exceeds the input voltage, confirming correct boost converter operation.

The simulation verified that the selected TPS61040 controller, 10 μ H inductor, and 47 μ F output capacitor were capable of achieving the required voltage conversion while maintaining stable operation. The results were used to validate component selections and confirm that the converter could satisfy the design objective of generating a regulated 12 V output from a 5 V input source.

5.4 Boost Converter Switching Node Voltage Validation

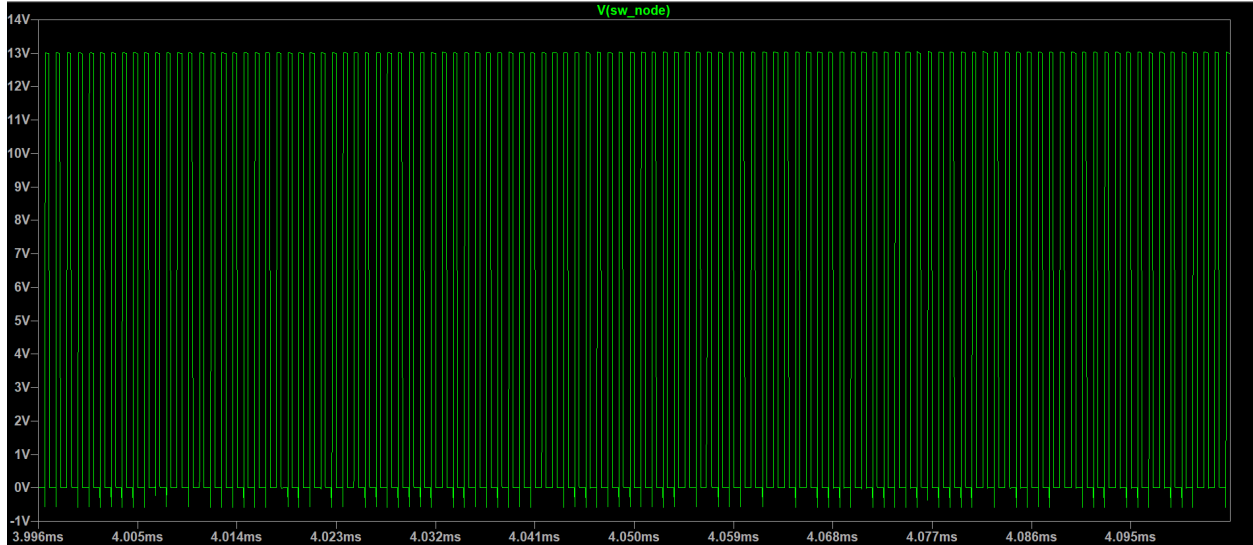


Figure 7. Boost Converter Switching Node Voltage

The switching node waveform was monitored to verify energy-transfer operation within the converter. The rapid voltage transitions observed at the switching node confirm the expected switching behavior of the boost topology. The waveform demonstrates the periodic storage and release of energy by the inductor and confirms proper interaction between the switching transistor, inductor, diode, and output capacitor.

5.5 Boost Inductor Current Validation

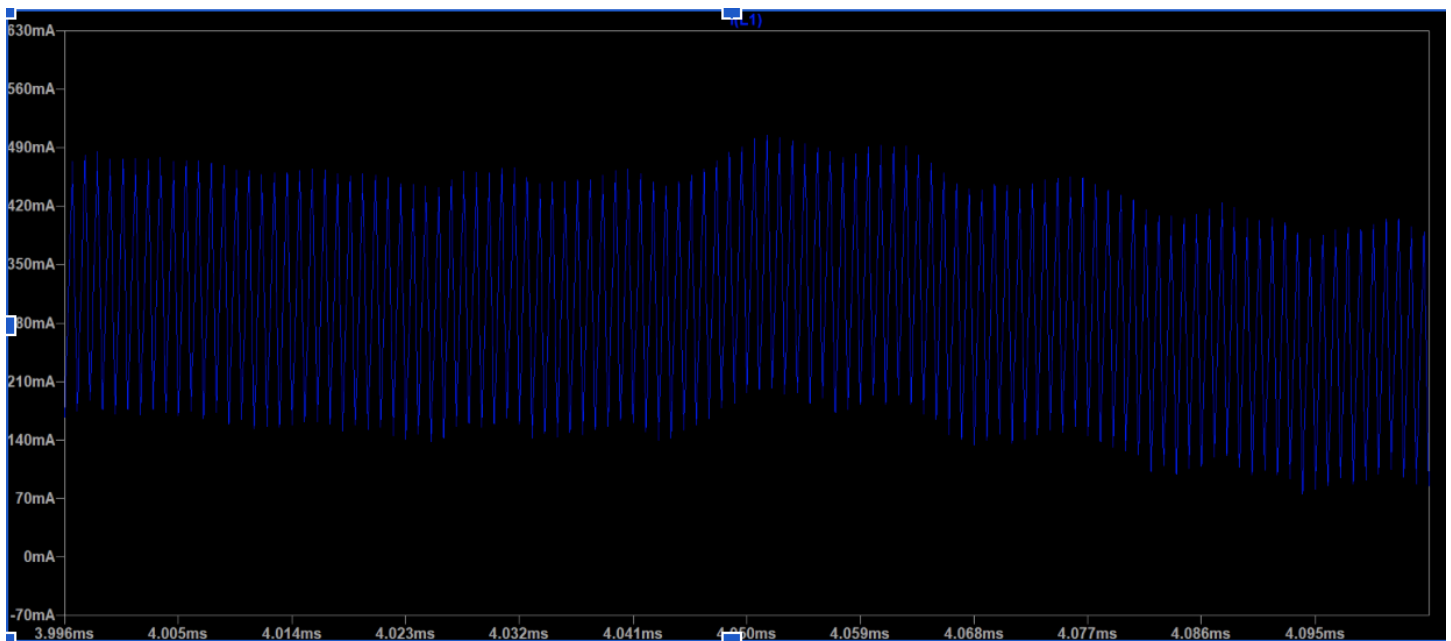


Figure 8. Inductor Ripple Current Waveform

The boost converter inductor current waveform was monitored to verify proper energy storage and transfer operation. The waveform exhibits the expected triangular ripple current characteristic of continuous conduction mode (CCM) boost converter operation. During the switch ON period, current increases as energy is stored within the magnetic field of the inductor. During the switch OFF period, the current decreases as the stored energy is transferred through the Schottky diode to the output capacitor and load.

5.6 Simulation Summary

The LTspice validation confirmed successful operation of both converter topologies prior to PCB implementation. The buck converter achieved regulated 5 V output from a 12 V input source while the boost converter achieved regulated 12 V output from a 5 V input source. Startup response, switching-node behavior, and output-voltage regulation were evaluated to verify correct converter operation and component sizing decisions. The simulation results were subsequently used to support inductor selection, output capacitor optimization, feedback network validation, and overall topology verification before schematic capture and PCB layout implementation in Altium Designer.

6. Final Schematic Design

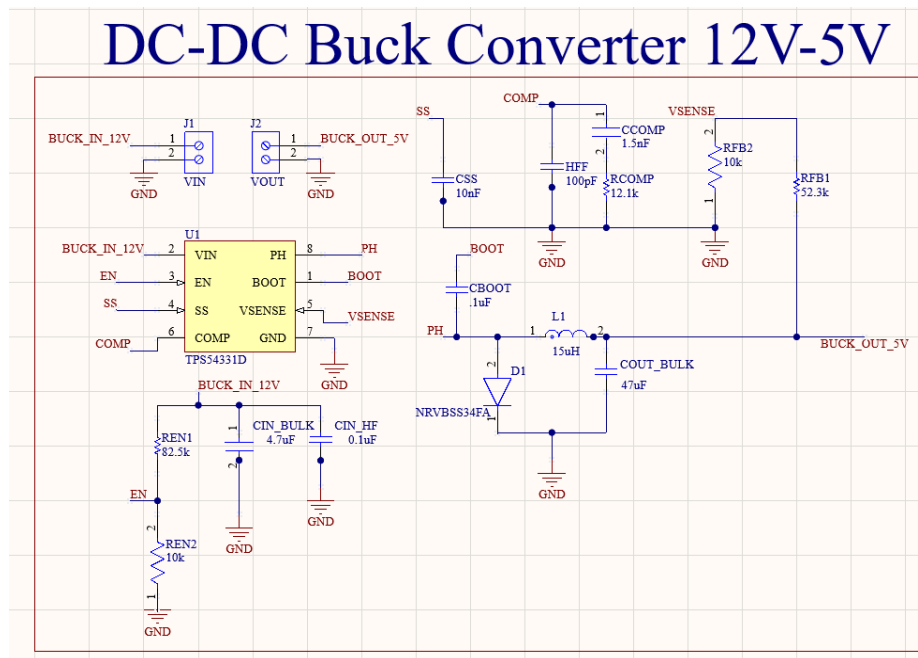


Figure 9. Buck Converter Schematic

Buck Converter Section

- TPS54331D controller
- 12V $\rightarrow$ 5V conversion
- Feedback network establishes output voltage
- Compensation network provides loop stability
- Input/output filtering minimizes ripple

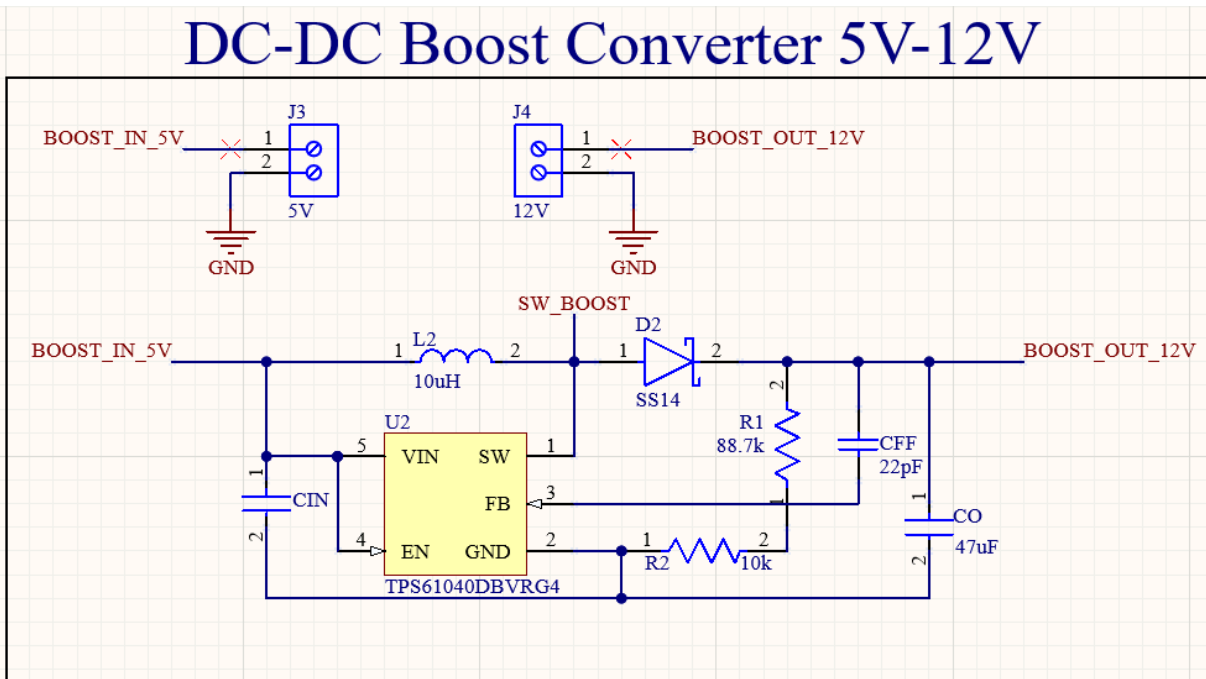


Figure 10. Boost Converter Schematic

Boost Converter Section

- TPS61040 controller
- 5V → 12V conversion
- Feedback divider regulates output voltage
- Schottky rectifier provides output energy transfer
- Output capacitor stabilizes voltage during switching

7.PCB Design Layout Considerations

The PCB layout was developed using Altium Designer while following common switching power supply layout practices. Particular attention was given to current-loop minimization, component placement, grounding strategy, and manufacturability.

7.1 Component Placement Strategy

The highest-current components were placed in close proximity to each controller IC to minimize parasitic resistance and inductance. Input capacitors, switching regulators, inductors, output capacitors, and Schottky diodes were arranged to create compact power-conversion stages while maintaining routing accessibility.

7.2 High-Current Path Routing

Switching converter current loops were kept as short as practical to reduce radiated EMI and voltage overshoot. Wider copper traces were used for power paths carrying input, output, and switching currents. Sensitive feedback traces were routed away from noisy switching nodes whenever possible.

7.3 Ground Plane Implementation

A continuous ground pour was implemented to provide a low-impedance return path and improve thermal performance. Ground planes also help reduce noise coupling and improve overall converter stability by minimizing return-current loop area.

7.4 Design Rule Verification

The PCB design was verified using Altium Design Rule Checks (DRC) and Electrical Rule Checks (ERC). Verification included:

- Clearance checking
- Trace width verification
- Unconnected net detection
- Component footprint validation
- Schematic-to-PCB synchronization

All identified issues were resolved prior to finalizing the design.

7.5 Testability Considerations

Test points were incorporated to allow access to critical nodes during future hardware validation. Key measurement locations include:

- Input voltage rails
- Output voltage rails
- Ground reference points
- Switching nodes

These locations would support oscilloscope and multimeter measurements during board bring-up and troubleshooting.

Design Note: The PCB was developed as a proof-of-concept design intended to demonstrate complete hardware-development workflow proficiency, including schematic capture, component selection, simulation-based validation, PCB implementation, and design-rule verification.

8. Conclusion

This project demonstrated the complete design workflow of a dual-function DC-DC converter board incorporating both buck and boost converter topologies. Engineering calculations were performed to establish duty-cycle requirements, size passive components, and design feedback networks capable of achieving the desired output voltages. Manufacturer-recommended components were selected and evaluated through LTspice simulation to validate converter operation, output regulation, switching behavior, and component performance. The design was subsequently implemented in Altium Designer through schematic capture, component selection, PCB development, and design-rule verification. The final design successfully achieved the project objectives of converting 12 VDC to 5 VDC using a buck converter and 5 VDC to 12 VDC using a boost converter while applying practical power-electronics design principles. Throughout the project, emphasis was placed on component selection methodology, simulation-based validation, switching-regulator design practices, PCB layout considerations, and engineering documentation.

This project provided valuable experience in power-supply design, hardware development workflows, and PCB implementation while demonstrating skills directly applicable to embedded hardware and electronics engineering roles. Future work would include PCB fabrication, hardware bring-up, efficiency characterization, thermal analysis, and oscilloscope-based validation of converter performance under varying load conditions.